Similarity-Calibrated Conformal Prediction: a physics-derived, a-priori coverage certificate for transfer across operating regimes

Theory note for internal review

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Status of this note. This is the formal core for review. The guarantee (Theorem 1) is *rigorous given* the modeling hypotheses H1–H2; H1–H2 are themselves falsifiable physical content, not theorems, and the diagnostic of §5 tests whether they are operative on given data.

1 Setup and notation

A population of units is indexed by operating condition $c \in \mathcal{C}$, each with known physical parameters θ_c (e.g. load P_c, speed n_c, temperature T_c). A unit under condition c, monitored at a given instant, yields features X and true remaining useful life Y, with joint law $\mathbb{P}_c$. A model trained on held-out data supplies a point predictor $\hat{y}(\cdot)$ and a nonconformity score $V = s(X, Y)$; we use the one-sided over-prediction score $V = \hat{y}(X) - Y$ throughout (a lower RUL bound). Write $\mathcal{V}_c$ for the law of V under $\mathbb{P}_c$, and $\mathrm{d_{TV}}$ for total variation distance.

Definition 1 (Dimensionless reduction)**.** From the governing physics, the Buckingham Π theorem yields a set of dimensionless groups. We partition them into (i) *scale-setting* groups, used to build a characteristic score scale $\sigma(\theta_c) > 0$ such that the *dimensionless score* $\tilde{V} = V/\sigma(\theta_c)$ has a condition-invariant law under exact similitude, and (ii) a *similitude-departure parameter* $\boldsymbol{\psi}(\theta_c) \in \mathbb{R}^m$ collecting any residual groups that the scale does not absorb (incompletely modeled or unmodeled physics). Write $\tilde{\mathcal{V}}_c$ for the law of $\tilde{V}$ under $\mathbb{P}_c$. By construction σ absorbs the dominant operating-parameter scaling; $\boldsymbol{\psi}$ is constant across $\mathcal{C}$ when the dimensionless model is complete.

Two instantiations. *(i) Rolling-contact fatigue* (Lundberg–Palmgren / ISO 281): rating life $\propto (C/P)^p$, $p = 3$; the scale $\sigma(\theta_c) = T_L(c) \propto (C/P)^p/n$ *absorbs* the load–speed group P/C, so large load/speed differences do not by themselves induce residual mismatch; $\boldsymbol{\psi}$ collects residual effects (e.g. lubrication regime, the ISO 281 a_{ISO} modifiers). *(ii) Catalyst deactivation* (first order, Arrhenius $k(T) = A\, e^{-E/RT}$): the scale $\sigma(\theta_c) = T_L(c) = 1/k_{\mathrm{nom}}(T_c)$ *contains* the primary rate group $e^{-E/RT}$, so the post-normalisation law is invariant under temperature alone; $\boldsymbol{\psi}$ captures unmodeled rate channels. In both, matching the scale-setting groups enforces dynamic similitude: the dimensionless trajectory is condition-invariant and conditions differ only through σ.

2 Hypotheses

Assumption 1 (Lipschitz similitude departure)**.** There exists $L < \infty$ such that $\mathrm{d_{TV}}\big(\tilde{\mathcal{V}}_c, \tilde{\mathcal{V}}_{c'}\big) \le L\, \|\boldsymbol{\psi}(\theta_c) - \boldsymbol{\psi}(\theta_{c'})\|$ for all $c, c' \in \mathcal{C}$.

The bound is on the *residual* dependence after scale absorption. The boundary case (exact similitude, "H1") is $\boldsymbol{\psi} \equiv \text{const} \Rightarrow \tilde{\mathcal{V}}_c = \tilde{\mathcal{V}}_{c'}$ for all c, c', *irrespective of how far apart the scale-setting parameters are*; the global Lipschitz form ("H2") is the regularity needed away from exact similitude. Crucially the certificate is *not* loosened by large differences in the dominant operating parameters (load, speed, temperature) —those are absorbed by σ— but only by departure from similitude, $\|\Delta\boldsymbol{\psi}\|$.

3 The SCC procedure

Split conformal performed in dimensionless coordinates:

1. Calibration $\{(X_i, Y_i)\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} \mathbb{P}_S$ (source); scores $\tilde{V}_i = s(X_i, Y_i)/\sigma(\theta_S)$.
2. Dimensionless quantile $\tilde{q} = \tilde{V}_{(\lceil(1-\alpha)(n+1)\rceil)}$ (order statistic of the $\tilde{V}_i$).
3. For a target unit under condition T, return, in original units, $\mathcal{C}(X_{n+1}) = \{y : \; s(X_{n+1}, y) \le \sigma(\theta_T)\,\tilde{q}\}$ (the quantile is re-dimensionalised by the *target* scale).

Naive conformal is the special case $\sigma \equiv 1$ (calibrate on raw V): it ignores the scale change across conditions and miscovers when $\sigma(\theta_S) \neq \sigma(\theta_T)$.

4 Coverage certificate

Lemma 1 (Barber et al., 2023; non-exchangeable coverage)**.** *For split conformal with weights w_i and pre-fitted scores, the coverage gap obeys* $(1-\alpha) - \mathbb{P}(Y_{n+1} \in \mathcal{C}) \le \big(1+\sum_{i=1}^n w_i\big)^{-1} \sum_{i=1}^n w_i \, \mathrm{d_{TV}}\big(R(Z), R(Z^i)\big)$, *where $R(Z)$ is the score vector and $R(Z^i)$ swaps test point with point i. For independent data,* $\mathrm{d_{TV}}(R(Z), R(Z^i)) \le 2\,\mathrm{d_{TV}}(Z_i, Z_{n+1})$.

Theorem 1 (SCC certificate)**.** *Under Assumption 1, with calibration i.i.d. from S and target $(X_{n+1}, Y_{n+1}) \sim \mathbb{P}_T$ independent of calibration,*

$$\mathbb{P}\big(Y_{n+1} \in \mathcal{C}(X_{n+1})\big) \;\ge\; 1 - \alpha - \frac{2n}{n+1}\, L \,\big\|\boldsymbol{\psi}(\theta_S) - \boldsymbol{\psi}(\theta_T)\big\| \;\ge\; 1 - \alpha - 2L\,\big\|\boldsymbol{\psi}(\theta_S) - \boldsymbol{\psi}(\theta_T)\big\|.$$

Proof. Coverage is the event $\{s(X_{n+1}, Y_{n+1}) \le \sigma(\theta_T)\tilde{q}\} = \{\tilde{V}_{n+1} \le \tilde{q}\}$ with $\tilde{V}_{n+1} = s(X_{n+1}, Y_{n+1})/\sigma(\theta_T) \sim \tilde{\mathcal{V}}_T$. Were $\tilde{V}_1, \ldots, \tilde{V}_{n+1}$ exchangeable (all $\tilde{\mathcal{V}}_S$), split conformal gives $\mathbb{P}(\tilde{V}_{n+1} \le \tilde{q}) \ge 1 - \alpha$. They are not: $\tilde{V}_{n+1} \sim \tilde{\mathcal{V}}_T \neq \tilde{\mathcal{V}}_S$. Apply Lemma 1 with $w_i \equiv 1$ (denominator $n+1$). The pre-fitted scores are independent scalars; swapping calibration point i with the test point exchanges $\tilde{V}_i \sim \tilde{\mathcal{V}}_S$ and $\tilde{V}_{n+1} \sim \tilde{\mathcal{V}}_T$, while the other $n-1$ coordinates are unchanged. The two joint laws are product measures differing only in those two independent coordinates; by subadditivity of TV over independent coordinates, $\mathrm{d_{TV}}(R(Z), R(Z^i)) \le \mathrm{d_{TV}}(\tilde{\mathcal{V}}_S, \tilde{\mathcal{V}}_T) + \mathrm{d_{TV}}(\tilde{\mathcal{V}}_T, \tilde{\mathcal{V}}_S) = 2\,\mathrm{d_{TV}}(\tilde{\mathcal{V}}_S, \tilde{\mathcal{V}}_T)$ for every i (the independent-data case of Lemma 1). Summing n identical terms, the gap is $\le \frac{2n}{n+1}\,\mathrm{d_{TV}}(\tilde{\mathcal{V}}_S, \tilde{\mathcal{V}}_T)$. Assumption 1 gives $\mathrm{d_{TV}}(\tilde{\mathcal{V}}_S, \tilde{\mathcal{V}}_T) \le L\|\boldsymbol{\psi}(\theta_S) - \boldsymbol{\psi}(\theta_T)\|$. □

Corollary 1 (A-priori certificate, two regimes)**.** $\|\boldsymbol{\psi}(\theta_S) - \boldsymbol{\psi}(\theta_T)\|$ *is a physical quantity, not a function of target failure data.* (a) Complete similitude *($\boldsymbol{\psi} \equiv$ const): the bound is* 0 *and target coverage is* $\ge 1-\alpha$ for arbitrary differences in the scale-setting parameters *(load, speed, temperature), requiring only the target scale $\sigma(\theta_T)$ from θ_T.* (b) Incomplete similitude*: coverage is certified a priori up to $2L\|\Delta\boldsymbol{\psi}\|$, given a bound on the departure (the magnitude of unmodeled physics). In both regimes no density ratio is estimated — the property that distinguishes SCC from weighted/shift conformal, whose coverage control requires likelihood ratios of $\mathbb{P}_T$ to $\mathbb{P}_S$ from target data. The diagnostic of §5 certifies which regime holds.*

Remark 1 (Estimating L and the departure). With $m \geq 2$ source conditions one estimates L and a finite-sample intercept a (the $\mathrm{d_{TV}}$ floor, $a \to 0$ as $n^{-1/2}$) by regressing the empirical $\mathrm{d_{TV}}(\tilde{\mathcal{V}}_S, \tilde{\mathcal{V}}_{S'})$ on $\|\Delta\boldsymbol{\psi}\|$ over source pairs; the operational bound $2(a + L\|\Delta\boldsymbol{\psi}\|)$ then upper-bounds the target gap. This uses only source data — it is a priori with respect to the target.

5 Similitude diagnostic: knowing when the certificate applies

Assumption 1 bites when $\boldsymbol{\psi}$ is non-constant across conditions (a governing group is unmodeled). Because exact similitude implies the dimensionless score (equivalently, for RUL, the dimensionless life $D = Y\,\sigma(\theta_c)^{-1}$) is condition-invariant, it is testable from source data alone. For each pair (c, c') form $g = \big[\mathrm{median}(D \mid c)/\mathrm{median}(D \mid c')\big]$; under similitude $g = 1$ (the bearing capacity / common scale constant cancels). A bootstrap CI on g yields a three-way verdict: 1 inside a tight CI $\Rightarrow$ *holds*; 1 outside $\Rightarrow$ *violated*; CI wider than a preset fold $\Rightarrow$ *indeterminate* (underpowered). On *indeterminate* or *violated* data, SCC declines the a-priori certificate and defers to data-driven (weighted) conformal. This makes SCC self-aware of its operating envelope rather than silently over-certifying.

6 Scope and limitations

(1) Assumptions H1–H2 are physical hypotheses; the guarantee is conditional on them. (2) The certificate bounds the gap for the *modeled* groups; unmodeled physics contributes a residual gap, which the controlled-departure study quantifies (the gap grows monotonically and remains under the operational bound) and the diagnostic detects. (3) Validation here is on controlled physical simulation, where similitude is genuine but not assumed by the estimator; demonstration on data whose degradation is not similitude-governed (and is underpowered, e.g. small bearing cohorts with large within-condition spread) returns *indeterminate* by design. (4) Real-process validation on adequately-powered field data is the principal item of future work.